TEXORA · LECTURE SHEET

System Protection and Earthing

How power systems defend themselves: the principles of protection for transformers and motors; the relay, circuit breaker, switches and starters that do the work; protection against electric shock and fire; and earthing — what it is, why it matters, and how it is done.

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 4 · LEVEL 2 TERM 2

1. Why protection is needed

Electrical faults — short circuits, earth faults, overloads, insulation failure — deliver enormous energy in milliseconds. Unchecked, a fault current ten or twenty times the normal current overheats conductors, starts fires, destroys machines and can kill. A **protective system** must detect the abnormal condition and isolate *only* the faulty part, as fast as possible, while the healthy remainder of the system keeps running. A good protection scheme has five qualities: **reliability** (it operates when it should, and only then), **selectivity** (it trips the smallest possible section), **speed** (damage grows with I^2t, so milliseconds matter), **sensitivity** (it detects even small fault currents), and **economy** (protection should cost a small fraction of what it protects).

2. Principles of protection for transformers

- **Overcurrent and earth-fault protection:** fuses (on small units) or overcurrent relays with circuit breakers on the HT side respond to short circuits and heavy overloads in or beyond the transformer.
- **Differential protection:** CTs on both sides compare the current entering with the current leaving; any difference must be flowing through a fault *inside* the transformer, so it trips instantly. Used on larger transformers.
- **Buchholz relay:** fitted in the pipe between the tank and the conservator of an oil-filled transformer. Incipient faults decompose the oil and generate gas, which the relay collects to give an alarm; a violent internal fault drives an oil surge that trips the breaker. It is the classic internal-fault guard of the oil-filled transformer.
- **Thermal protection:** winding and oil temperature indicators and relays alarm or trip on overheating from sustained overload or cooling failure.

- **Surge protection:** lightning arresters (Topic 3) divert voltage surges to earth before they puncture the winding insulation.

3. Principles of protection for motors

- **Overload protection:** a thermal relay (heated bimetal) or electronic overload relay in the starter trips the motor when current stays moderately above rating for too long — the signature of mechanical overloading or a jammed load. It is deliberately slow to allow normal starting current.
- **Short-circuit protection:** fuses or a magnetic circuit breaker (MCCB) clear the very large currents of a short circuit almost instantly; they do not protect against small overloads, so they complement, never replace, the overload relay.
- **Single-phasing protection:** if one supply phase fails, a three-phase motor keeps running on two phases and draws destructive current in the healthy windings; a phase-failure relay or the overload relay's differential action trips it.
- **Earth-fault protection:** leakage relays detect current escaping to the frame or earth and disconnect the motor before the frame becomes live and dangerous.
- **Under-voltage protection:** the starter's contactor drops out when supply fails, so the motor cannot restart unexpectedly when power returns — a vital safety feature around machinery.

4. The protective devices

DEVICE	WHAT IT DOES
Fuse	A calibrated weak link that melts and breaks the circuit when current exceeds its rating for long enough. Simple, fast and cheap, but it is a one-shot device: after operation it must be replaced. Protects against overcurrent and short circuits.
Protective relay	The "brain" of the scheme: it continuously measures current (through CTs), voltage (through PTs) or temperature, and when the quantity crosses its setting it closes a trip circuit that opens the breaker. Types include overcurrent, earth-fault, differential, Buchholz and thermal relays. A relay draws almost no power itself.
Circuit breaker	The "muscle": a switch built to make and break load and fault currents without damage, its contacts separating in an arc-quenching medium (air, oil, vacuum or SF_6 gas). Common forms: MCB (miniature, for final circuits), MCCB (moulded-case, for distribution), ACB (air circuit breaker, for LT mains) and vacuum or SF_6 breakers at HT. It is opened manually or by a relay's trip coil.
Switch / isolator	Provides on-load switching (switch) or a visible off-load safety gap (isolator). Neither clears faults by itself (Topic 3).

Starter	The motor’s combined controller: a contactor for switching, an overload relay for protection, and start/stop push buttons. Direct-on-line (DOL) starters suit small motors; star–delta starters reduce the starting current of larger ones by first connecting the windings in star.
RCD / ELCB	Residual-current (earth-leakage) device: compares outgoing and returning currents and trips within milliseconds when the difference — current leaking through a person or faulty insulation — exceeds a small threshold such as 30 mA. The main personal-protection device in final circuits.

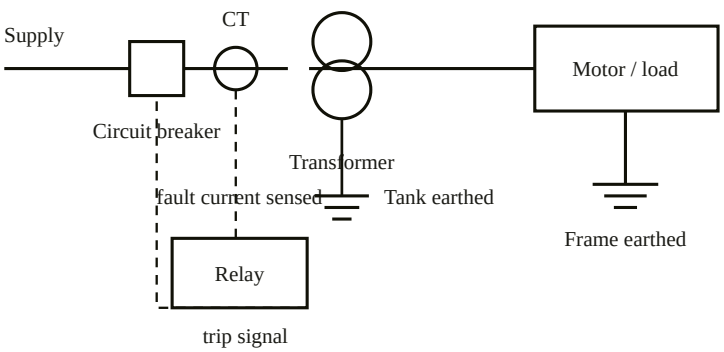


Fig. 1 — A simple protection scheme: the CT senses the fault, the relay decides, the breaker’s trip coil disconnects — and exposed metalwork is earthed.

5. Protection against electric shock and fire

Electric shock occurs when a person bridges a live part and earth (or two live parts) and current passes through the body; currents of a few tens of milliamperes through the chest can be fatal. Protection is layered:

- **Insulation and enclosure** of live parts, with safe clearances and heights, so contact cannot happen accidentally.
- **Earthing of all exposed metalwork**, so a fault to the frame becomes a large fault current that trips the protection instantly, instead of leaving the frame silently live (Section 6).
- **Automatic disconnection** by fuses, breakers and especially RCDs, which limit both the size and the duration of any shock current.
- **Extra-low voltage** (SELV) for hand lamps and tools in wet or confined places, and double-insulated appliances that need no earth.

Electrical fires start from overloaded cables run beyond their rating, loose connections heating at a point, short circuits, and arcs in damaged insulation. Protection is correct cable sizing with matching

fuse or breaker ratings, tight and inspected terminations, earth-fault protection to catch insulation failure early, and keeping flammable material (in a mill: cotton fly and lint) away from electrical equipment.

6. Earthing: what it is and why it matters

Earthing (grounding) is the connection of a point of an electrical system, or of exposed metalwork, to the general mass of the earth by a low-resistance conductor. Its purposes are:

- **Personal safety:** if a live conductor touches a motor's frame, the earth path carries a large fault current that trips the breaker at once; without earthing the frame would sit at full voltage, waiting for a victim to complete the circuit.
- **Voltage stability:** earthing the neutral of a supply system pins the system voltages to earth potential and prevents dangerous over-voltages from lightning, surges or accidental contact with a higher-voltage line.
- **A path for lightning and fault currents** to disperse safely into the ground.

Two kinds are distinguished: **system earthing** — earthing a live point of the system, normally the neutral of a generator or transformer — and **equipment earthing** — earthing metal parts that do not normally carry current, such as frames, tanks and enclosures. The quality of an earth connection is its **earth resistance**. It must be low, because the fault current (and hence the speed of tripping) and the voltage appearing on the frame both depend on it: in practice, values of about 1 Ω or less are aimed for at substations, and a few ohms (commonly up to about 5 Ω) are accepted for domestic and small installations.

7. Types of earthing

TYPE	CONSTRUCTION	REMARKS
Plate earthing	A copper plate (about 60 cm × 60 cm × 3 mm) or galvanised-iron plate (about 6 mm thick) buried vertically about 3 m deep, the earth wire bolted to it; the pit is filled with alternate layers of charcoal and salt around the plate, with a watering pipe and funnel to keep the soil moist.	The standard substation and building earth; low resistance and long life.
Pipe earthing	A perforated galvanised-iron pipe (about 38 mm diameter, 2–2.5 m long) driven or buried vertically in permanently moist	Cheap and widely used for houses and small installations.

	ground; the earth wire is clamped inside its top.	
Rod earthing	A solid copper or steel rod driven deep into the ground (often several rods in parallel, spaced apart).	Useful in sandy or rocky soil where pits are difficult; driving reaches moist layers.
Strip / wire earthing	Bare copper wire or strip buried in a shallow horizontal trench.	Used in rocky ground and for large substation earth grids, where many strips form a buried mesh under the whole yard.

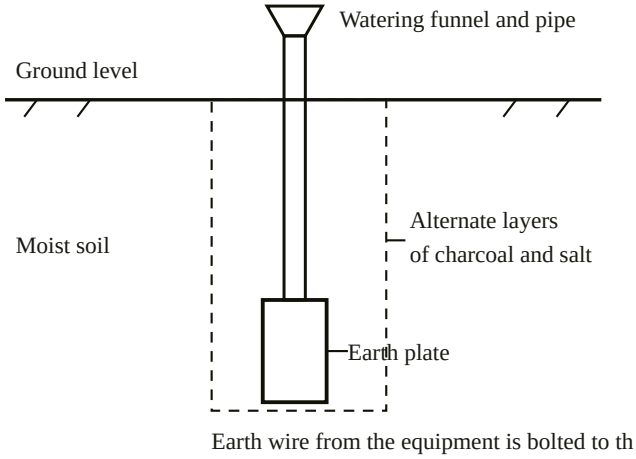


Fig. 2 — Plate earthing: the electrode pit, with the watering pipe that keeps the contact resistance low.

Whatever the type, the earth lead must be mechanically strong, corrosion-resistant and continuous — an earth that has corroded away is worse than none, because it gives confidence without protection. Periodic measurement of earth resistance (by an earth tester) is part of substation maintenance (Topic 3).

8. Key terms

- **Fault:** any abnormal condition — short circuit, earth fault, overload, open phase — that protection exists to detect and isolate.
- **Selectivity:** the property of tripping only the faulty section, leaving healthy sections in service.
- **Relay:** a measuring and decision device that orders a breaker to trip when its setting is exceeded.
- **Buchholz relay:** a gas- and surge-operated relay protecting oil-filled transformers against internal faults.
- **Starter:** a motor controller combining a contactor, overload relay and control buttons (DOL, star–delta).

- **RCD:** residual-current device; trips on a small imbalance between outgoing and returning current, protecting people against shock.
- **Earth resistance:** the resistance between an earth electrode and the general mass of the earth; the lower, the better the protection.

9. Key points

- Protection = sensing (CTs, PTs, relays) + decision (relay settings) + interruption (circuit breaker), applied selectively and fast.
- Transformers: overcurrent, differential and Buchholz protection, plus temperature monitoring. Motors: overload (thermal), short circuit (fuse/MCCB), single-phasing and earth-fault protection, grouped in the starter.
- Fuses melt, relays decide, breakers interrupt, starters control motors, RCDs protect people.
- Shock protection layers: insulation, earthing of metalwork, automatic disconnection, RCDs, and extra-low voltage where risk is high.
- Earthing serves safety, voltage stability and surge discharge. System earthing grounds the neutral; equipment earthing grounds the frames.
- Earth resistance must be low and maintained — plate earthing with charcoal and salt layers and watering is the standard substation method.

10. Summary

A power system protects itself in layers. Instrument transformers scale the fault down to measurable size; relays compare it against their settings; breakers interrupt it; starters give every motor its own overload, short-circuit and phase-failure guardian; and earthing turns a dangerous silent fault on a machine frame into a large, instantly tripped fault current, while holding the system’s voltages steady against earth. The scheme of Fig. 1 — sense, decide, trip, earth — is the pattern repeated, at every scale, from a house’s 30 mA RCD to a grid substation’s differential relays.

11. Practice

Answer in your own words; brief answers follow. No multiple-choice questions.

1. A factory motor keeps tripping its overload relay a few minutes after starting, though the fuse never blows. Explain what each observation tells you, and name the first thing you would check.
2. Why is the Buchholz relay fitted only on oil-filled transformers, and what two stages of action does it provide?

3. A worker receives a shock from a machine frame but the breaker does not trip. Give two possible defects in the installation that together explain this.
4. An earth electrode measures 40 Ω. Why is this unacceptable, and name two measures used to reduce earth resistance.
5. Explain why a 30 mA RCD protects a person but is not, by itself, protection against a short circuit between live and neutral.

Answer key

1. The overload relay tripping after minutes indicates a sustained moderate overcurrent — an overload, not a short circuit; the fuse stays intact because the current is far below its fast-operation level. First check the driven load for mechanical overload or jamming, then check for single phasing and for a wrongly set overload relay.
2. It works by collecting gas from, and detecting oil surges in, the insulating oil, so it can exist only where the transformer is oil-immersed. Stage one: gas from a small, developing fault collects and operates an alarm. Stage two: a violent fault drives an oil surge that trips the circuit breaker immediately.
3. (Any two:) the frame is not earthed (earth wire broken or never connected), so the fault current is limited to the path through the worker; the earth resistance is far too high, limiting fault current below the breaker's trip threshold; the breaker or its trip circuit is defective or wrongly rated, so it does not operate at the fault level reached.
4. At 40 Ω, a fault to a frame would pass only a small current (for 230 V, under 6 A), too little to trip most breakers, while the frame would rise to nearly full voltage — the protection fails exactly when it is needed. Reduce it by enlarging or deepening the electrode (plate, multiple rods in parallel) and by keeping the soil around it moist and low-resistivity (charcoal and salt layers, watering, or a better site).
5. An RCD compares the currents in the live and neutral conductors and trips on a difference of 30 mA — the leakage through a person to earth. In a live–neutral short circuit the two currents remain equal (both enormous), so the RCD sees no imbalance; only the fuse or breaker, responding to the magnitude of the current, clears that fault.
